PWE in plain words

What it is, why we use it, and what to say when someone asks

The one sentence to remember

PWE is a fast, approximate version of what COMSOL does. It takes a hole shape and gives back the band frequencies. We used it to make thousands of training examples cheaply, then corrected its small error with real COMSOL runs.

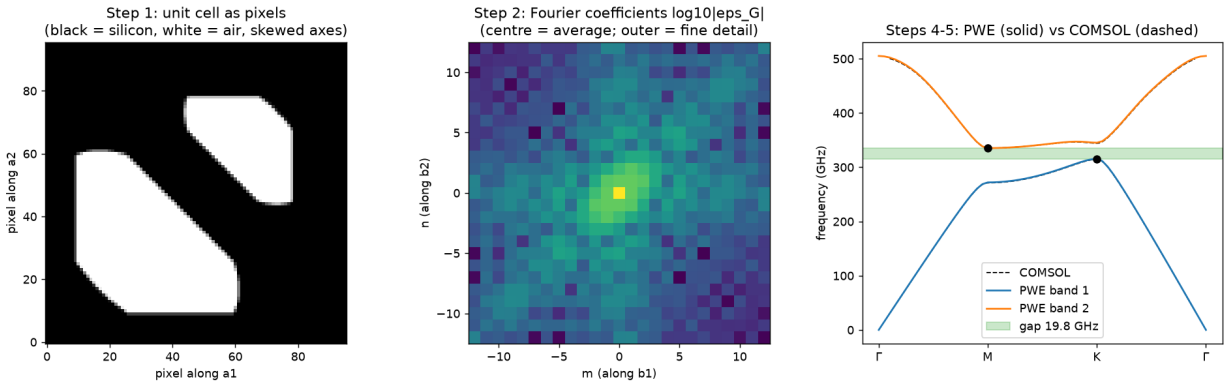
1. What problem does it solve?

A guitar string can only ring at certain notes, set by its length and tension. Light inside your photonic crystal is the same: it can only travel at certain frequencies, set by the pattern of holes. Between the allowed frequencies there is a range where light cannot travel at all. That range is the **band gap**, and its two edges (f_lower and f_upper) are the numbers the whole project is about.

Finding those allowed frequencies for a given hole shape is called **solving for the band structure**. COMSOL does it. PWE does exactly the same job, a different way.

2. How it works, in three steps (no maths)

1. **Draw the cell as a picture.** The computer makes a pixel image of one unit cell: silicon is black, the air holes are white (left panel below).
2. **Describe the picture as a mix of simple waves.** Just as a song can be written as a mix of pure notes, the hole pattern is written as a mix of simple ripple patterns (middle panel). This is the 'plane waves' in the name *plane-wave expansion*.
3. **Ask which light waves fit.** With the crystal written as waves, finding the allowed light frequencies becomes one table of numbers (a matrix) that the computer solves in a fraction of a second. Out come the bands (right panel), the same curves COMSOL gives.



Left: the unit cell as pixels. Middle: how strongly each simple wave is present (bright = strong). Right: the bands PWE finds (solid) on top of COMSOL's (dashed). They overlap; the green strip is the band gap.

COMSOL vs PWE in one line each

COMSOL cuts the cell into thousands of tiny triangles (the mesh) and solves on those. Accurate, flexible, slow.
PWE describes the cell as a few hundred simple waves and solves on those. Very fast, and accurate for a simple 2D cell like ours.
Both are standard textbook methods. PWE is the method in the well-known free MIT program MPB and in the textbook by Joannopoulos et al., *Photonic Crystals: Molding the Flow of Light* (2008).

3. Why we used it

	COMSOL	PWE (our code)
time per shape	much longer (you run batches of ~200 overnight / for hours)	about 0.17 s
shapes we have	4,198 (triangles, circles, freeform)	6,000
accuracy	the reference	0.79 / 0.34 GHz from COMSOL (3,267 cells)

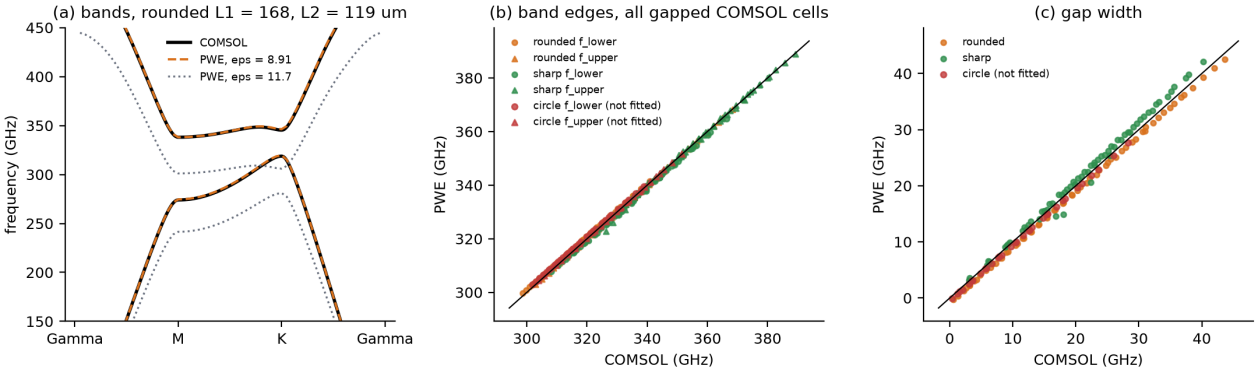
The AI needs thousands of examples to learn how 10 shape numbers change the band gap. Making 6,000 in COMSOL would take weeks. PWE made them in about 20 minutes.

We tested this directly (study guide section 9.1). Scored on 1007 freeform shapes simulated in COMSOL:

approach	error vs COMSOL (GHz)
best AI trained only on COMSOL data	0.32 / 0.23
PWE alone	0.45 / 0.43
PWE + a correction learned from COMSOL (what we use)	**0.15 / 0.46**

So each tool does what it is good at: PWE gives **quantity**, COMSOL gives **accuracy**, and the correction combines them. This is called **multi-fidelity** learning: cheap low-accuracy data plus a little expensive high-accuracy data.

4. How do we know PWE is right?



(a) Full bands for one cell: COMSOL (black) and PWE (orange dashes) lie on top of each other. (b), (c) Every gapped COMSOL cell: PWE (up) vs COMSOL (across). Points on the diagonal line mean they agree.

- 3,267 gapped COMSOL cells of four shape families: band edges within 0.79 / 0.34 GHz of COMSOL on average.
- It was tuned on only 128 triangle cells, so all the others (circles, freeform shapes, the new triangles) are an independent test.
- For scale: the gap itself is up to about 40 GHz wide and sits near 300 to 350 GHz, so 0.5 GHz is about 0.15 percent.

The one setting we tuned (in case someone asks about 11.7)

Silicon's textbook permittivity is 11.7. With 11.7, PWE's bands come out too low (grey dotted line in panel (a)). With 8.91 they match COMSOL exactly. The likely reason is that the COMSOL model is 2D and uses an *effective* value for a thin silicon slab (refractive index about 2.99 instead of 3.42). Shalaev et al. (Nature Nanotechnology, 2019) used 2.965 in the same kind of 2D model. We fitted this one number on the triangles only; circles and freeform shapes then matched without any extra tuning. (Check the material value in your COMSOL model so you can say it confidently.)

5. Questions your guide may ask, with answers

Q: What is PWE?

A: A plane-wave expansion solver: a standard, fast way to compute the band structure of a periodic structure. It does the same job as COMSOL's eigenfrequency study, using waves instead of a mesh.

Q: Why not just use COMSOL for everything?

A: Speed. The AI needs thousands of examples; PWE makes one in 0.2 s. I also tested a COMSOL-only model on 1,007 freeform COMSOL shapes: best 0.32 / 0.23 GHz, against 0.21 / 0.30 GHz for PWE plus the COMSOL correction.

Q: How do you know PWE is accurate?

A: I compared it with COMSOL on 3,267 gapped cells of four shape families: 0.79 / 0.34 GHz on average, and the gap widths follow COMSOL closely.

Q: Are 439 plane waves enough?

A: I re-solved 225 COMSOL cells with 177 to 1,011 plane waves. At 439 the edges are within ~0.5 GHz of the finest setting, and since the permittivity was calibrated at 439, the agreement with COMSOL (~0.5 GHz) is as good as with 1,011 waves, which are 5 times slower.

Q: Why permittivity 8.91 and not 11.7?

A: The COMSOL model is 2D, so it uses an effective index for the slab. 8.91 (n = 2.99) is the value that matches COMSOL; Shalaev et al. 2019 used n = 2.965 in a similar 2D model. I fitted it once on triangles and it transferred to circles and freeform shapes.

Q: Did you write the solver?

A: It is part of my project code (src/shapes/pwe.py), written with AI coding help. The method itself is standard and well documented (Joannopoulos et al.; the MIT program MPB uses the same idea). My contribution is the AI design pipeline that uses it, and checking it against COMSOL.

Q: What are its limits?

A: 2D, one polarisation, effective index: a small error (~0.5 GHz), so final designs are checked in COMSOL and the error is corrected with COMSOL data.

Q: Is the final design checked in COMSOL?

A: Yes. 14 AI designs for four targets were simulated in COMSOL; 12 hit both edges within 1 GHz (worst 1.05 GHz).

Q: How is accuracy measured?

A: In GHz on shapes the model never saw. A design's error is the worse of its two edges after re-simulation; below 1 GHz counts as success. Percentages would hide errors at 300 GHz.

6. A 60-second explanation you can say

Say it like this

"To train the AI I need thousands of examples of hole shapes and their band gaps. COMSOL is accurate but too slow for that, so I used a plane-wave expansion solver. It computes the same bands in a fraction of a second by describing the crystal as a sum of simple waves. I checked it against COMSOL on 3,267 cells: it agrees to under a GHz. Then I used my COMSOL runs to learn a small correction, which brings the error down to about 0.14 / 0.23 GHz. Finally, 12 of 14 AI designs simulated in COMSOL hit their target within 1 GHz."